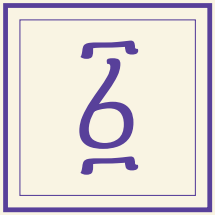
Computing clusters

The cost of computation and supercomputers

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 13 November 2026 𓃵𓃵𓃵𓃵



Section 1

One operation, and who does it

A FLOP is one operation on two numbers

One multiply accumulate

A FLOP is one operation on two numbers

One multiply accumulate

$a \times b$
one multiply

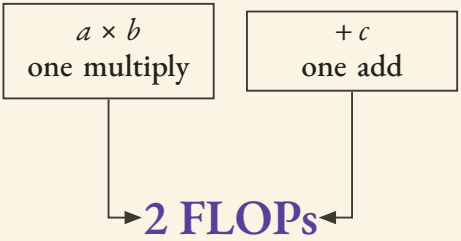
A FLOP is one operation on two numbers

One multiply accumulate

$a \times b$	$+ c$
one multiply	one add

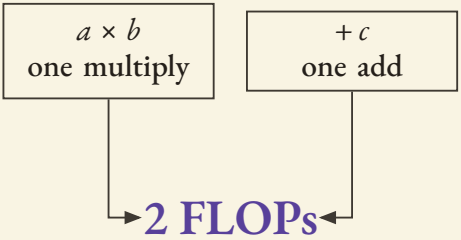
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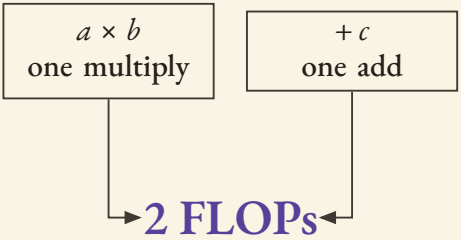


Count, or rate

FLOP: a count of operations
how much work

A FLOP is one operation on two numbers

One multiply accumulate



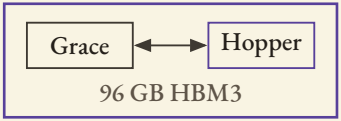
Count, or rate

FLOP: a count of operations
how much work

FLOP/s: operations per second
how fast

One computer, then and now



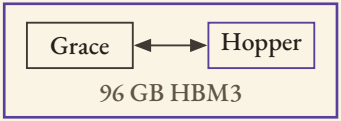


	Human computer, 1949	One GH200, 2024

Photograph: a computer room, NACA, 1949 (NASA Dryden, E49-54), public domain.

One computer, then and now



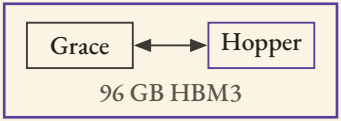


	Human computer, 1949	One GH200, 2024
Rate	1,000 additions a day	990×10^{12} FLOP/s

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One computer, then and now



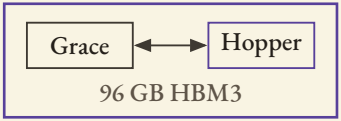


	Human computer, 1949	One GH200, 2024
Rate	1,000 additions a day	990×10^{12} FLOP/s
Format	Decimal, by hand	16 bits per number

Photograph: a computer room, NACA, 1949 (NASA Dryden, E49-54), public domain.

One computer, then and now



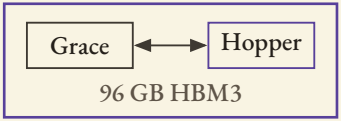


	Human computer, 1949	One GH200, 2024
Rate	1,000 additions a day	990×10^{12} FLOP/s
Format	Decimal, by hand	16 bits per number
Power	About 100 W	660 W on Alps

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One computer, then and now



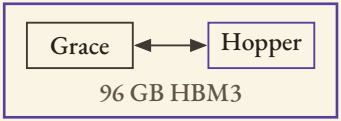


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Food	**2,065 kcal a day** *what one person eats*	**13,600 kcal a day** *what six or seven people eat*

Photograph: a computer room, NACA, 1949 (NASA Dryden, E49-54), public domain.

One computer, then and now





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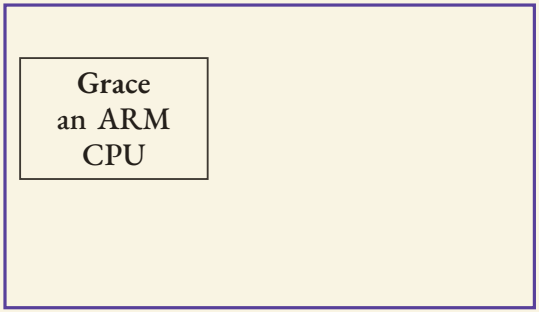
The clerk's rate is what the shop *delivered*. The chip's is what it *could* do.

Photograph: a computer room, NACA, 1949 (NASA Dryden, E49-54), public domain.

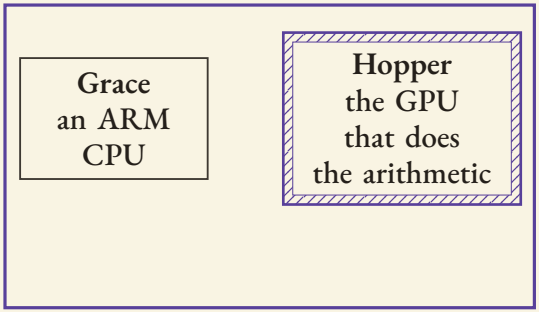
A GH200 is a CPU and a GPU on one module



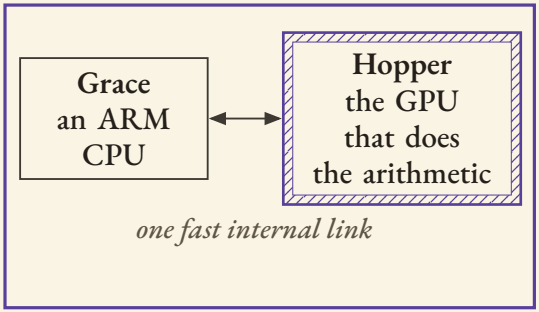
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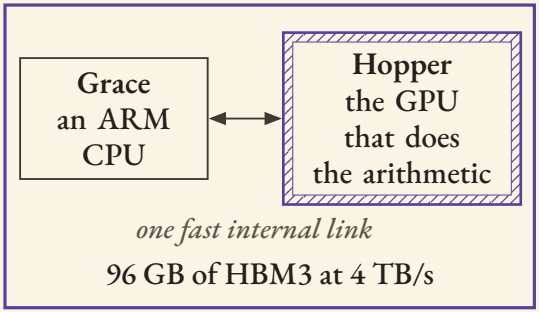
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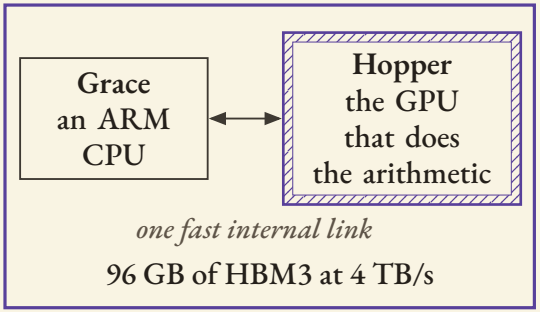
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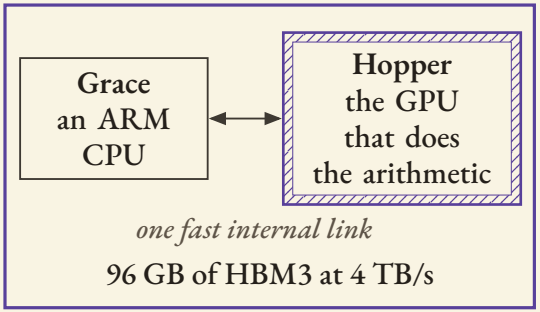


A GH200 is a CPU and a GPU on one module



The GPU computes. The CPU reads the data, runs Python, and keeps it fed

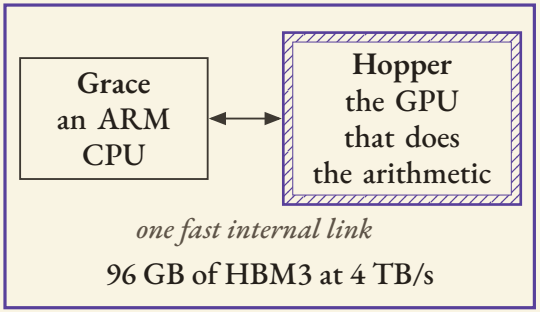
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The GPU computes. The CPU reads the data, runs Python, and keeps it fed

HBM3 is memory stacked on the GPU package: only 96 GB, but 4 TB/s

A GH200 is a CPU and a GPU on one module



- *The GPU computes. The CPU reads the data, runs Python, and keeps it fed*
- *HBM3 is memory stacked on the GPU package: only 96 GB, but 4 TB/s*
- *Between them 900 GB/s, about seven times what a plug-in card gets*

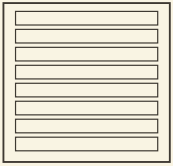
Alps is an HPE Cray EX254n

The TOP500 entry reads: *Alps, HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS*

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HPE Cray EX, the platform

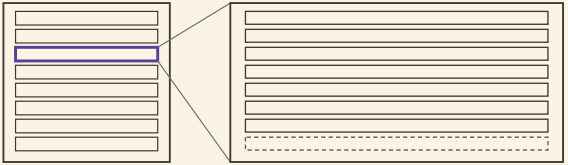


one of 24 cabinets: 8 chassis

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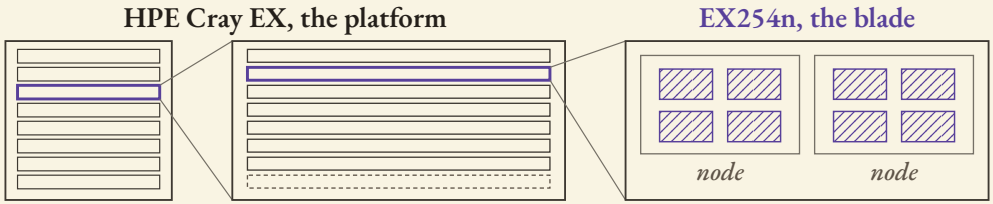


one of 24 cabinets: 8 chassis

one chassis: 8 blade slots, 7 used at Alps, for more power per GPU

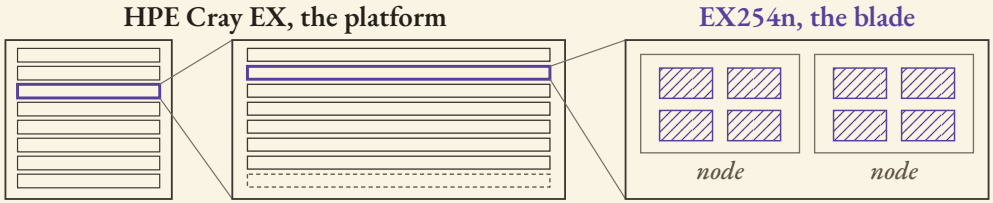
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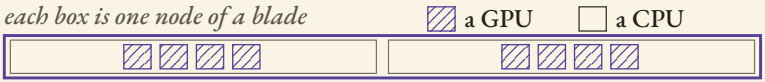
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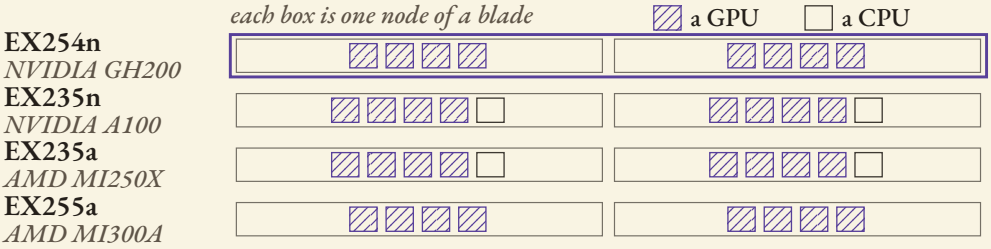
The platform: cabinets, liquid cooling, power and the network.

One platform, several blade types

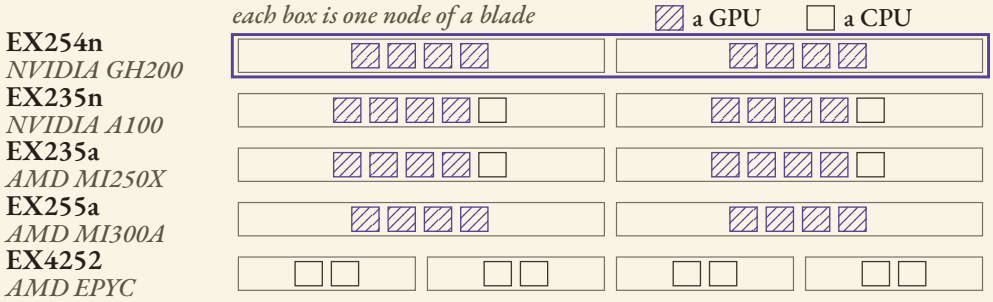
EX254n
NVIDIA GH200



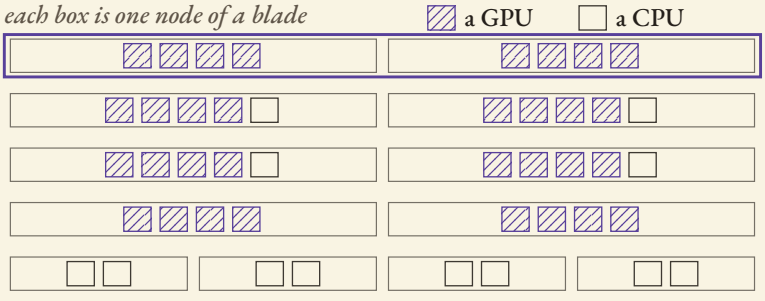
One platform, several blade types



One platform, several blade types

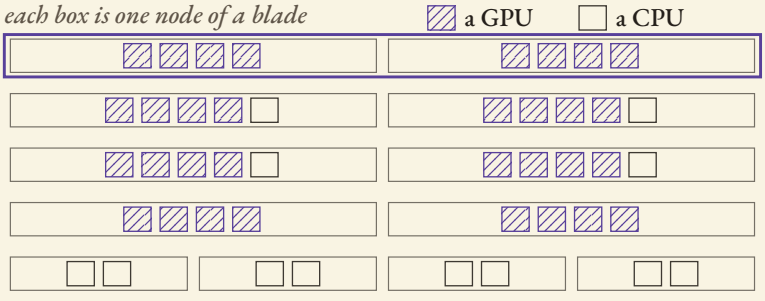


One platform, several blade types



Every name here that ends in n carries NVIDIA GPUs.

One platform, several blade types



Every name here that ends in n carries NVIDIA GPUs.
The blade choice is what makes Alps a machine for AI rather than a conventional CPU cluster.

The atelier against Alps

A collection of computers	Rate	Power
Prony's atelier, 1795 twenty people	0.694 additions/s	about 2 kW

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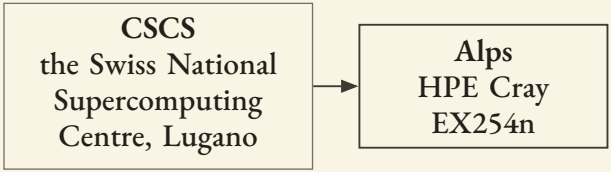
Fifteen billion billion times the rate,
on three and a half thousand times the power

CSCS and Alps, the machine behind Apertus

CSCS
the Swiss National Supercomputing Centre, Lugano

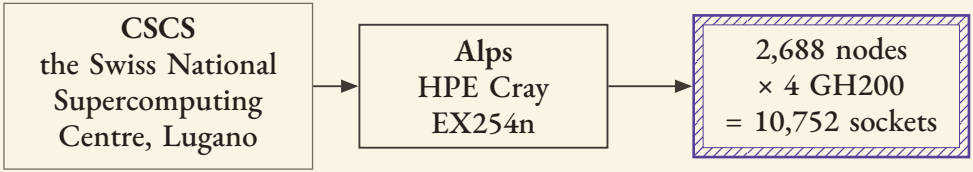
Machine: cscs.ch/computers/alps and top500.org/system/180259. Model: arXiv 2509.14233.

CSCS and Alps, the machine behind Apertus



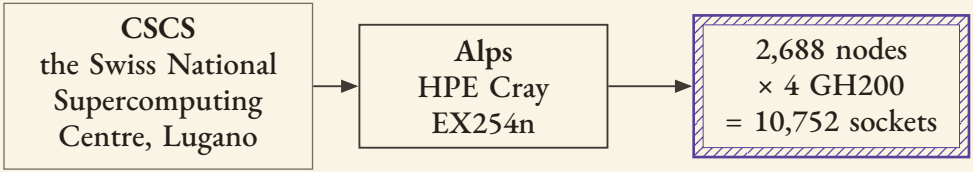
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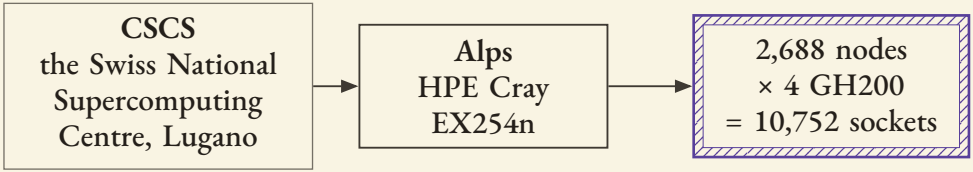


10.6 EFLOP/s $= 10.6 \times 10^{18}$ **FLOP/s**

all 10,752 GH200 together

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CSCS and Alps, the machine behind Apertus



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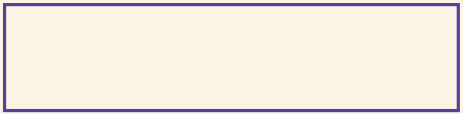
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Apertus, the open language model of EPFL, ETH Zurich and CSCS, was trained on Alps.

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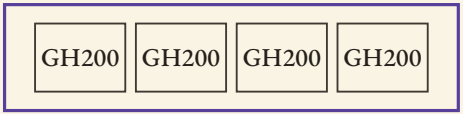
Nodes and sockets

One node



Nodes and sockets

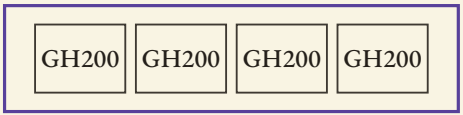
One node



One operating system, one address space

Nodes and sockets

One node



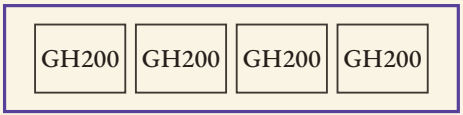
One operating system, one address space

Socket one processor module

Node one independent computer

Nodes and sockets

One node



One operating system, one address space

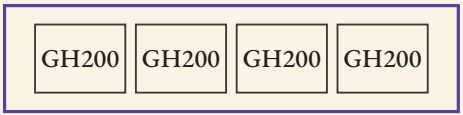
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2,688 nodes × 4 modules = 10,752 sockets

Nodes and sockets

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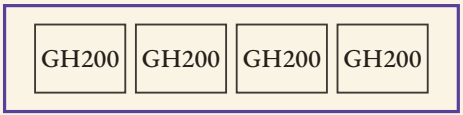
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Nodes and sockets

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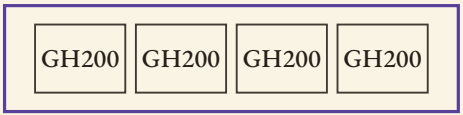
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Nodes and sockets

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2,688 nodes × 4 modules = 10,752 sockets

Socket to socket, inside the node: **150 GB/s**

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The node boundary is where the bandwidth falls off.

One second of a GH200, in three registers

The same one second of arithmetic	Time
The mean age of all 8.2 billion people alive, one addition each	8.3 microseconds

ResNet-50: He and others, arXiv 1512.03385: 3.8 billion multiply-adds, so 7.6×10^9 *FLOP per image. A peak count.*

One second of a GH200, in three registers

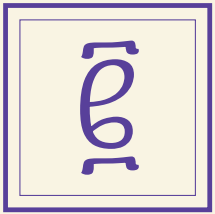
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Prony's atelier doing that same one second of work	45 to 136 million years

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Section 2

What the headline rate hides

Capacity and bandwidth are different numbers

96 GB
how much fits

Capacity and bandwidth are different numbers

96 GB
how much fits

4 TB/s
how fast you can move it

Capacity and bandwidth are different numbers

96 GB
how much fits

4 TB/s
how fast you can move it

Decides whether the weights and the optimiser state fit on one device at all

Capacity and bandwidth are different numbers

96 GB
how much fits

Decides whether the weights and the optimiser state fit on one device at all

4 TB/s
how fast you can move it

Decides how quickly the arithmetic units can be fed once the data is there

Capacity and bandwidth are different numbers

96 GB
how much fits

Decides whether the weights and the optimiser state fit on one device at all

4 TB/s
how fast you can move it

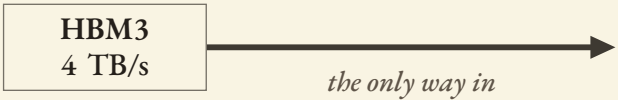
Decides how quickly the arithmetic units can be fed once the data is there

96 GB is less than an ordinary server holds. That is the trade: HBM is stacked on the package and wired short and wide, which is what buys the speed.

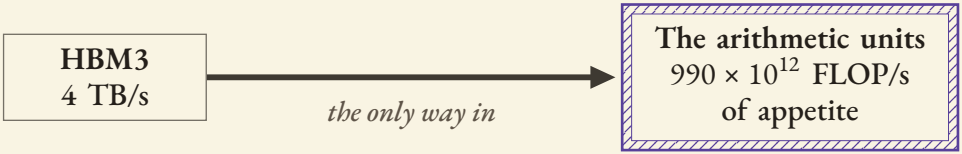
Two hundred and forty six operations per byte

HBM3
4 TB/s

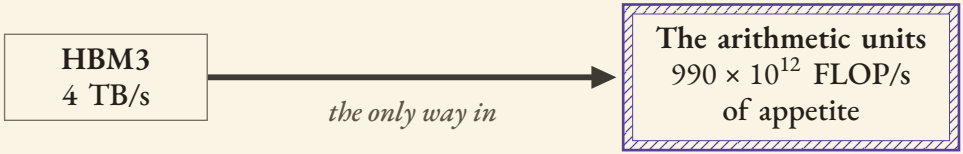
Two hundred and forty six operations per byte



Two hundred and forty six operations per byte

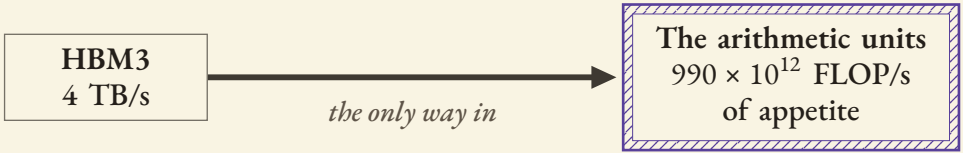


Two hundred and forty six operations per byte



$$\frac{990 \times 10^{12}\ \text{FLOP/s}}{4.023 \times 10^{12}\ \text{bytes/s}} = 246\ \text{FLOP per byte}$$

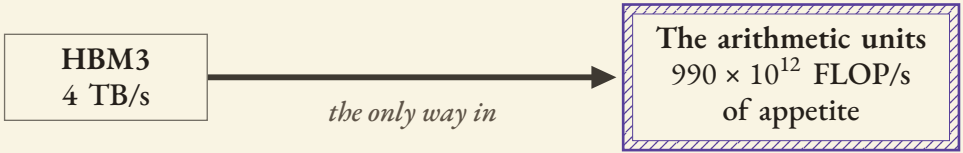
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A rate over a rate: the seconds cancel, and what is left is operations per byte

Two hundred and forty six operations per byte

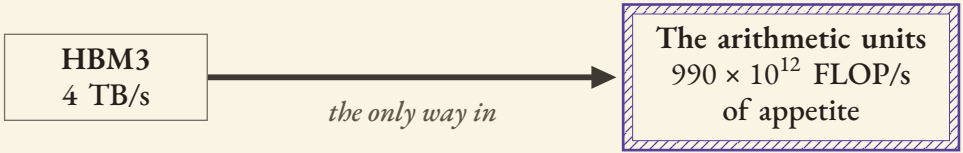


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A rate over a rate: the seconds cancel, and what is left is operations per byte

So unless the computation reuses each byte about that many times, the units wait. A large matrix multiply does reuse. Many other operations do not.

Two hundred and forty six operations per byte

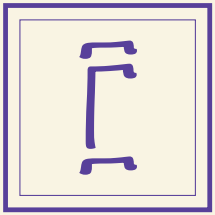


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A rate over a rate: the seconds cancel, and what is left is operations per byte

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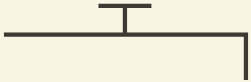
This is one of the reasons a real run reaches 21 to 46 percent of peak.



Section 3

The machine, and what it draws

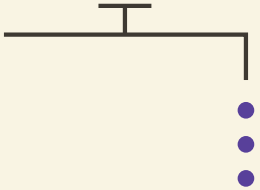
A watt is a rate, a kilowatt hour is an amount



660 W

How fast energy arrives

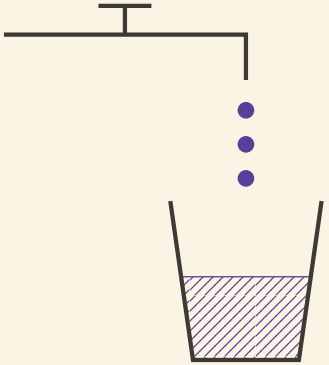
A watt is a rate, a kilowatt hour is an amount



660 W

How fast energy arrives

A watt is a rate, a kilowatt hour is an amount



660 W

How fast energy arrives

0.66 kWh *after one hour*

How much arrived

Watch the slash

km/h The slash divides, so it is a speed.

Watch the slash

km/h	The slash divides, so it is a speed.
kWh	No slash. The hour **multiplies**, so it is an amount.

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Longer gives you *more*, so the hour cannot be dividing.

Watch the slash

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kWh No slash. The hour **multiplies**, so it is an amount.

Longer gives you *more*, so the hour cannot be dividing.

---

rate × time = amount

Watch the slash

km/h The slash divides, so it is a speed.

kWh No slash. The hour **multiplies**, so it is an amount.

Longer gives you *more*, so the hour cannot be dividing.

rate	×	time	=	amount
FLOP/s	×	seconds	=	FLOP

Watch the slash

km/h The slash divides, so it is a speed.

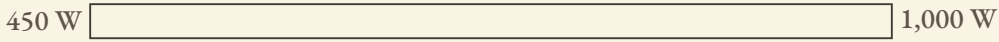
kWh No slash. The hour **multiplies**, so it is an amount.

Longer gives you *more*, so the hour cannot be dividing.

rate	×	time	=	amount
FLOP/s	×	seconds	=	FLOP
watts	×	hours	=	watt hours

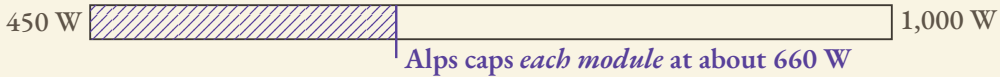
A wattage becomes a bill in two multiplications

One module is just a band



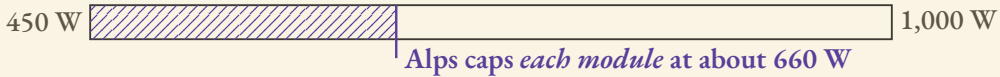
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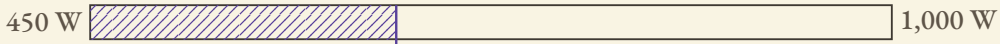
One module is just a band



The operator chooses: near 1,000 W is faster, lower saves energy and cooling

A wattage becomes a bill in two multiplications

One module is just a band



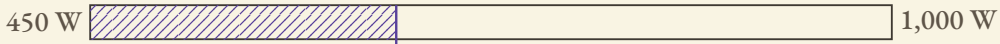
Alps caps *each module* at about 660 W

The operator chooses: near 1,000 W is faster, lower saves energy and cooling

660 W
for one hour

A wattage becomes a bill in two multiplications

One module is just a band



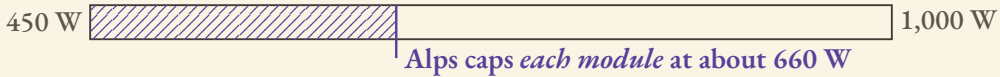
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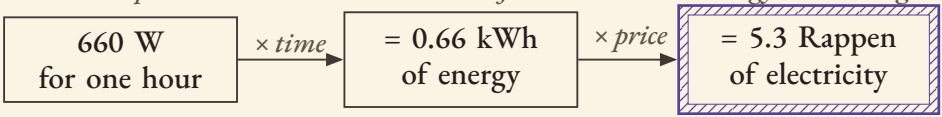
660 W for one hour → *× time* → = 0.66 kWh of energy

A wattage becomes a bill in two multiplications

One module is just a band



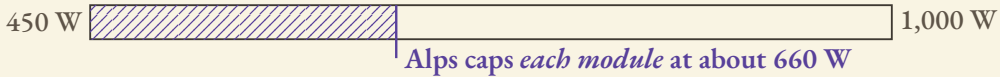
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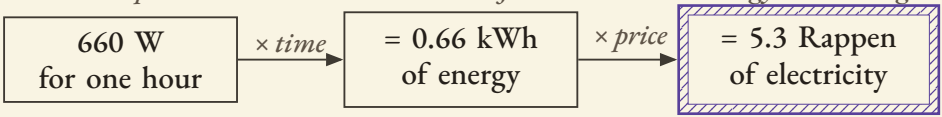
450 to 1,000 W: GH200 datasheet. 660 W: the Alps cap, docs.cscs.ch/running/slurm. 8 Rappen per kWh assumed (ElCom 2026, large consumers 6.75 to 9; households pay 27.7). Check: 10,752 × 660 W is 7,096 kW, against the 7,124 kW the TOP500 measured.

A wattage becomes a bill in two multiplications

One module is just a band



The operator chooses: near 1,000 W is faster, lower saves energy and cooling

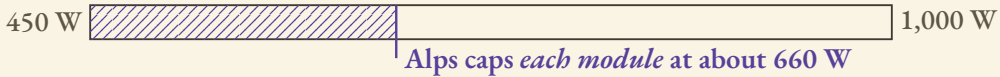


That is one module. All 10,752 is 7.1 megawatts, about 570 francs an hour.

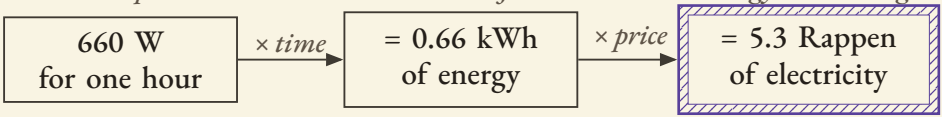
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One module is just a band



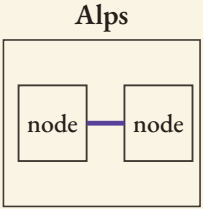
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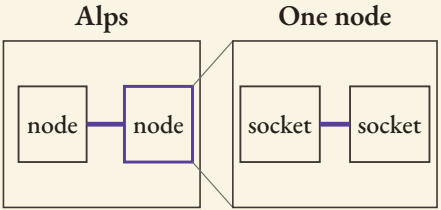
That is one module. All 10,752 is 7.1 megawatts, about 570 francs an hour.
The whole Apertus run is 3.96 GWh, about 0.3 million francs.

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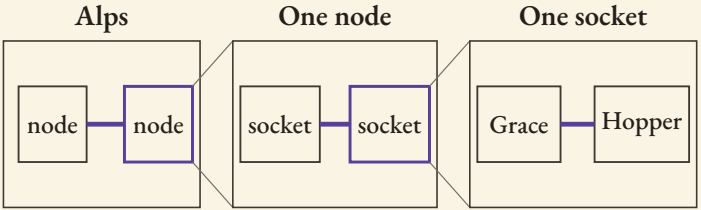
Four links, and each step down costs you



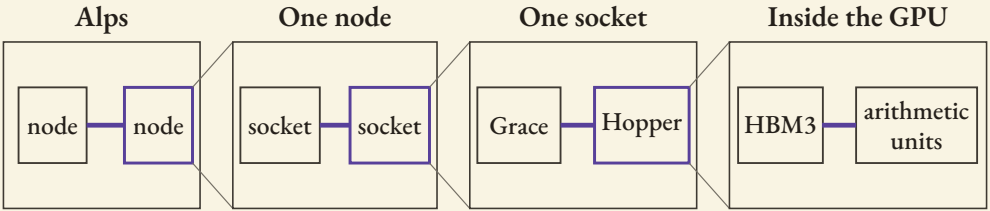
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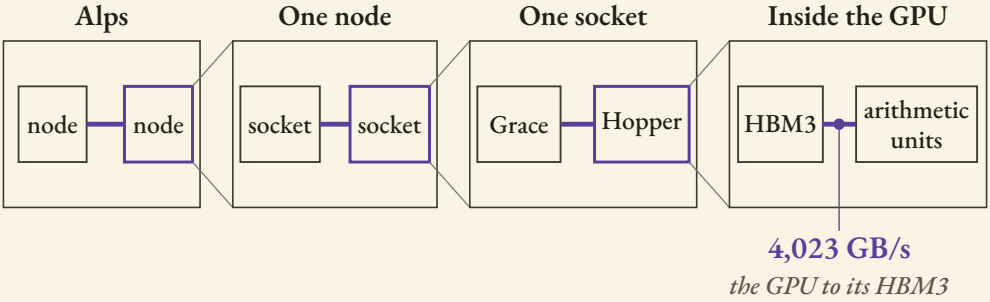
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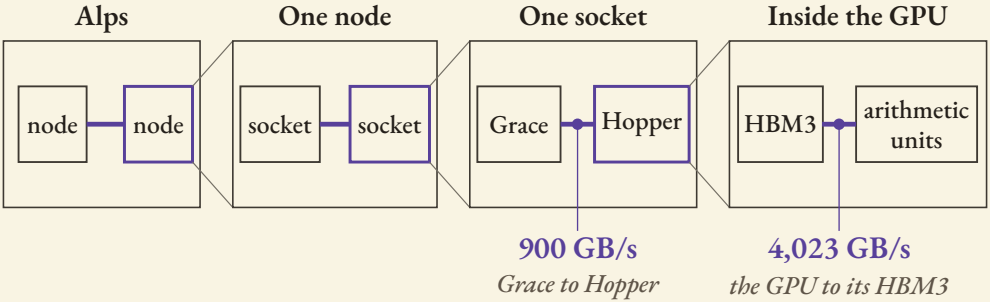
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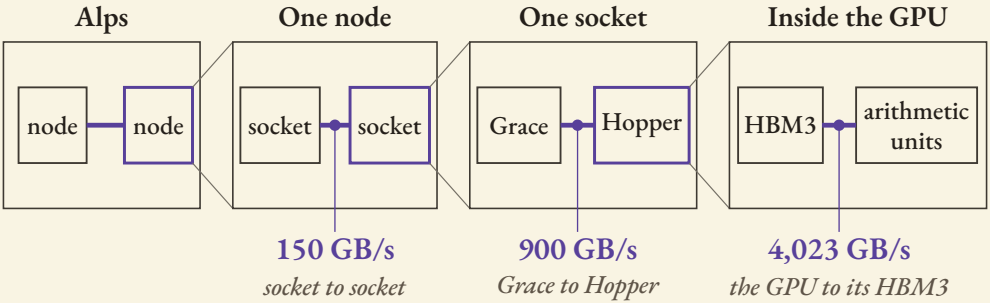
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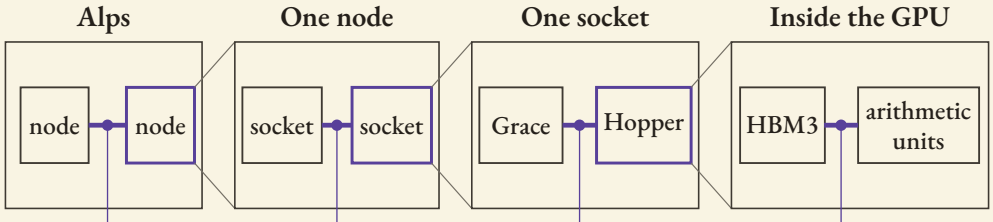
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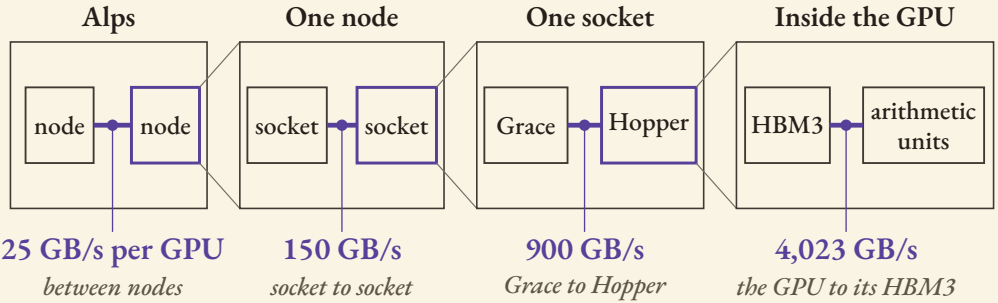
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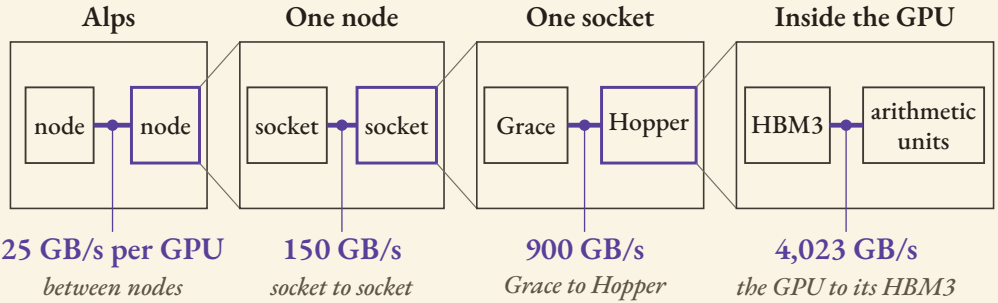


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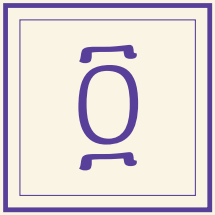
Each step down costs about a factor of five. End to end, 161 times.

Four links, and each step down costs you



Each step down costs about a factor of five. End to end, 161 times.

So you keep the data at the fast end and cross the slow links as rarely as you can.



Section 4

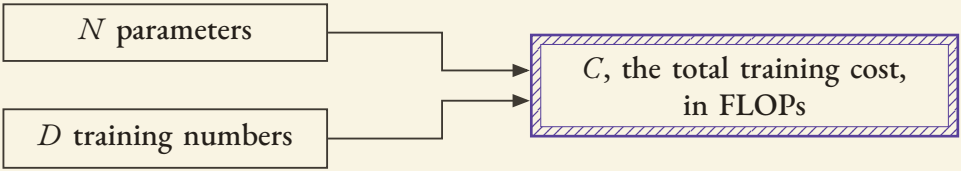
From a model to GPU hours

Two numbers price a training run

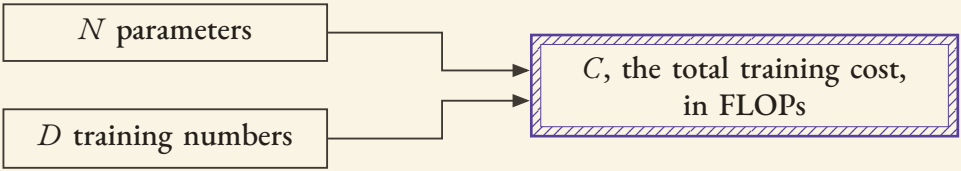
N parameters

D training numbers

Two numbers price a training run

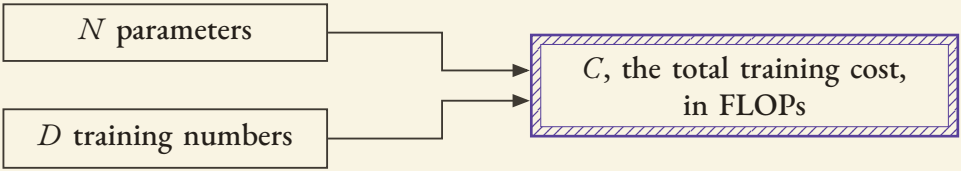


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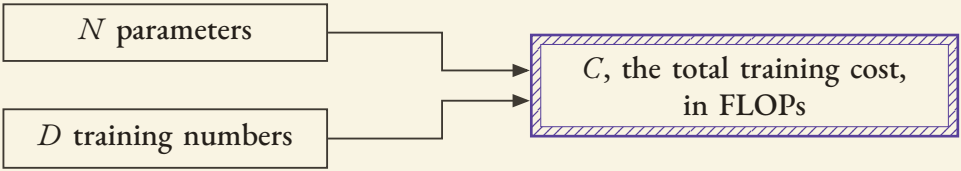
$$C = \underbrace{N}_{\text{parameters}}$$

Two numbers price a training run



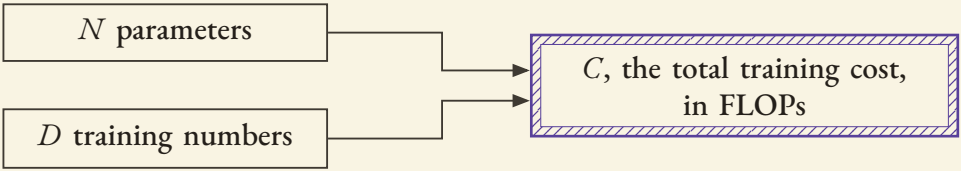
$$C = \underbrace{N}_{\text{parameters}} \times \underbrace{D}_{\text{training numbers}}$$

Two numbers price a training run



$$C = \underbrace{6}_{\text{per parameter, per training number}} \times \underbrace{N}_{\text{parameters}} \times \underbrace{D}_{\text{training numbers}}$$

Two numbers price a training run



$$C = \underbrace{6}_{\text{per parameter, per training number}} \times \underbrace{N}_{\text{parameters}} \times \underbrace{D}_{\text{training numbers}}$$

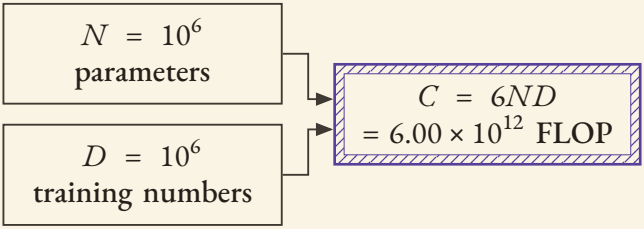
Kaplan and others, 2020, section 2.1. Here the 6 is a rule of thumb.

A million by a million is two minutes

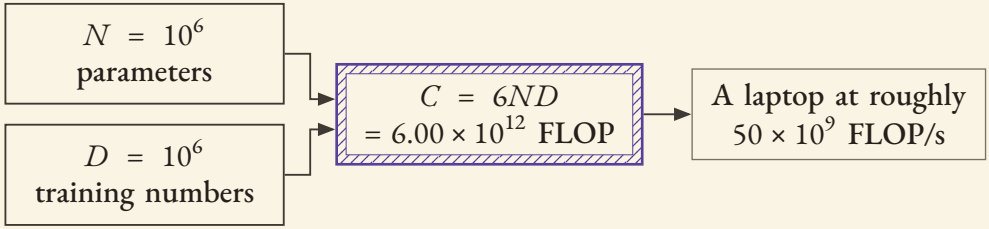
$N = 10^6$
parameters

$D = 10^6$
training numbers

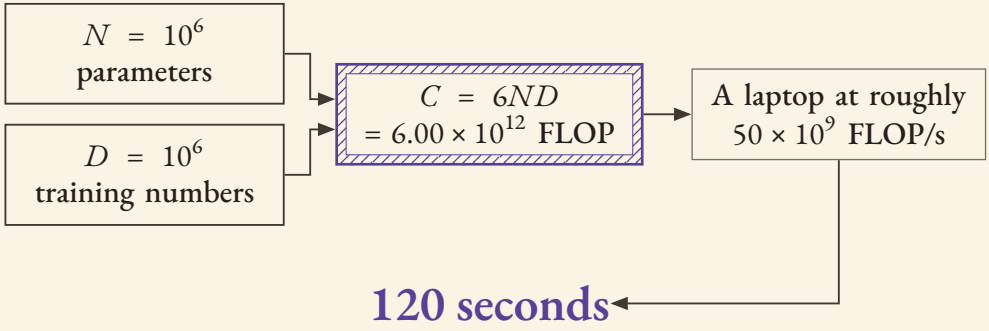
A million by a million is two minutes



A million by a million is two minutes



A million by a million is two minutes



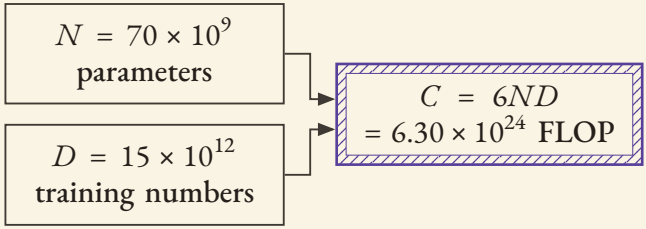
The same formula on Apertus-70B

$N = 70 \times 10^9$
parameters

$D = 15 \times 10^{12}$
training numbers

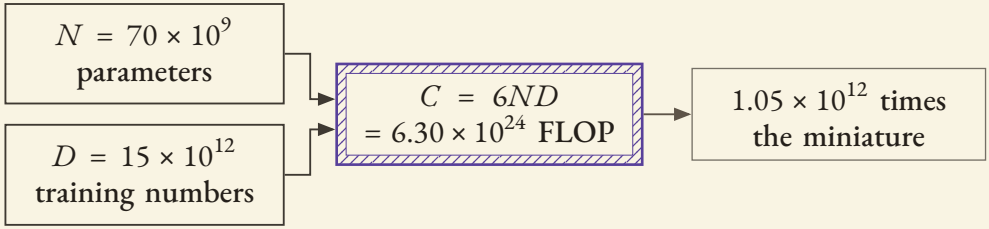
Model: arXiv 2509.14233.

The same formula on Apertus-70B



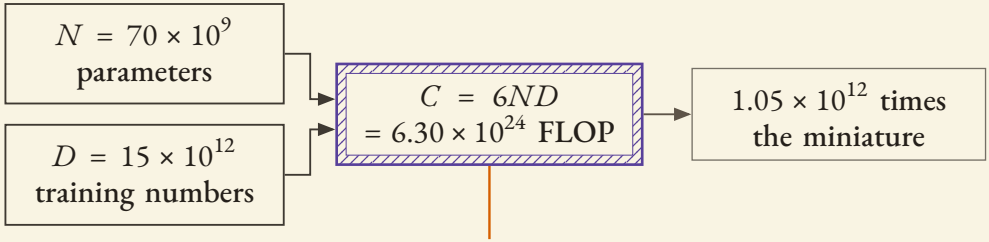
Model: arXiv 2509.14233.

The same formula on Apertus-70B



Model: arXiv 2509.14233.

The same formula on Apertus-70B



Model: arXiv 2509.14233.

From one GPU hour to the whole run

The run: training Apertus-70B on 15 trillion training numbers took over 6 million GPU hours, on up to 4,096 GPUs.

Run: arXiv 2604.12973.

From one GPU hour to the whole run

The run: training Apertus-70B on 15 trillion training numbers took over 6 million GPU hours, on up to 4,096 GPUs.

4,096 GPUs flat out
for 61 days

Run: arXiv 2604.12973.

From one GPU hour to the whole run

The run: training Apertus-70B on 15 trillion training numbers took over 6 million GPU hours, on up to 4,096 GPUs.

4,096 GPUs flat out for 61 days	It spanned about 91 days, so 2,747 on average

Run: arXiv 2604.12973.

From one GPU hour to the whole run

The run: training Apertus-70B on 15 trillion training numbers took over 6 million GPU hours, on up to 4,096 GPUs.

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The gap is ramp up, restarts and smaller phases: a campaign rarely runs at full width without a break

Run: arXiv 2604.12973.

From one GPU hour to the whole run

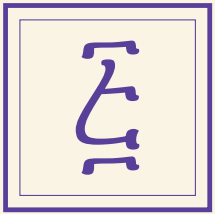
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4,096 GPUs flat out for **61 days**	It spanned about 91 days, so **2,747** on average

The gap is ramp up, restarts and smaller phases: a campaign rarely runs at full width without a break

6 million GPU hours × 660 W = **3.96 GWh**
× 8 Rappen = **0.32 million francs**

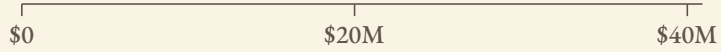
Run: arXiv 2604.12973.



Section 5

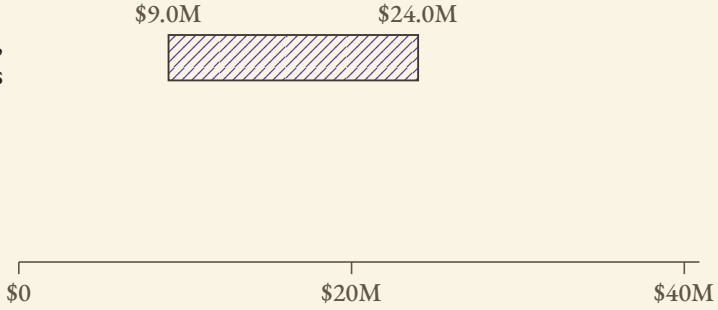
Dollars and energy

Six million hours, priced at cloud rates

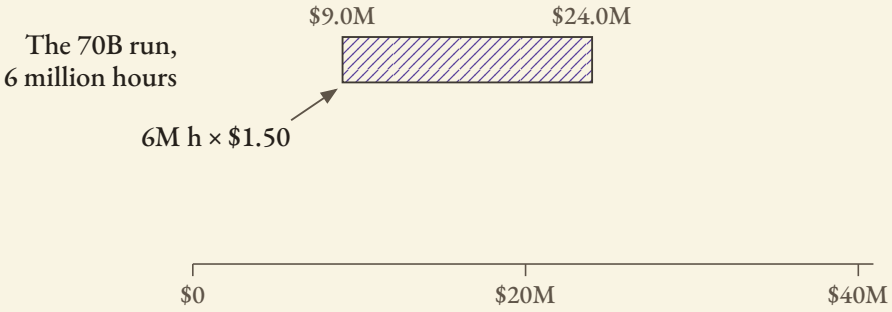


Six million hours, priced at cloud rates

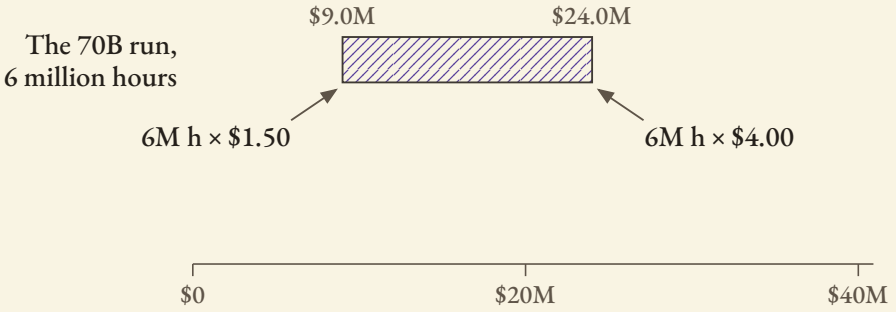
The 70B run,
6 million hours



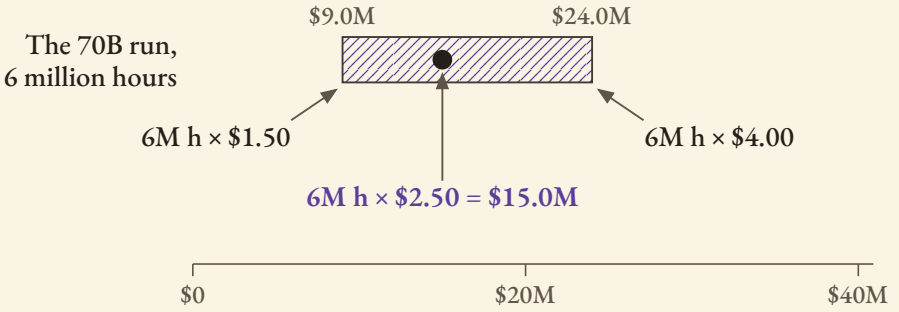
Six million hours, priced at cloud rates



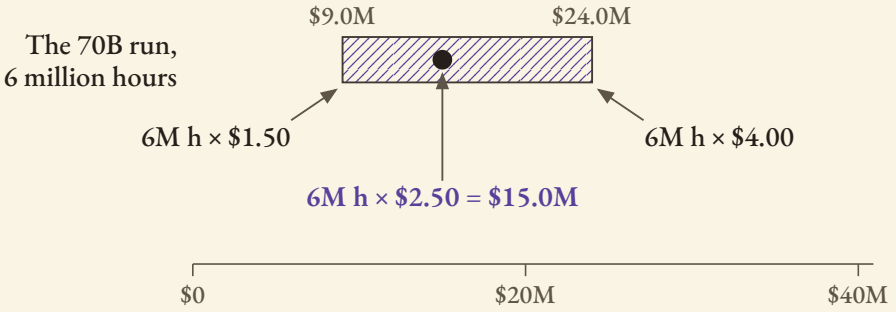
Six million hours, priced at cloud rates



Six million hours, priced at cloud rates



Six million hours, priced at cloud rates



The $1.50 to $4.00 band is an H100 rental survey, November 2025.

The same hours, in gigawatt hours

8.85 million GPU hours
90 days × 4,096 GPUs

The same hours, in gigawatt hours

8.85 million GPU hours
90 days × 4,096 GPUs

×

560 W
per module

The same hours, in gigawatt hours

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90 days × 4,096 GPUs

×

560 W
per module

=

4.95 GWh

The same hours, in gigawatt hours

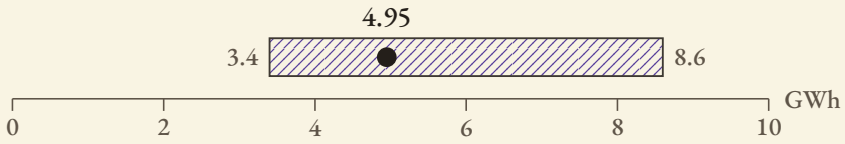
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90 days × 4,096 GPUs

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The same hours, in gigawatt hours

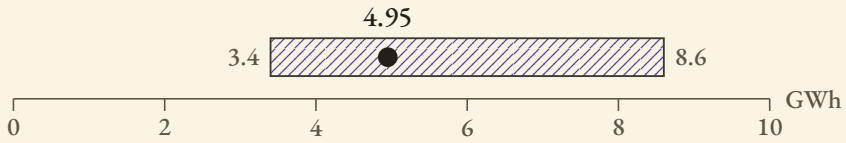
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About 1,100 Swiss households for a year, at 4,500 kWh each.

The same hours, in gigawatt hours

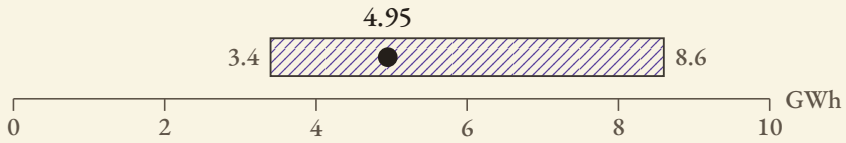
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90 days × 4,096 GPUs

×

560 W
per module

=

4.95 GWh



About 1,100 Swiss households for a year, at 4,500 kWh each.

Range 750 to 1,900 households, over four sourced module figures: 560 W, tech report, arXiv 2509.14233; 660 W, CSCS cap, docs.cscs.ch/running/slurm; 855 W, from the TOP500 run; datasheet band 450 to 1,000 W. Household 4,500 kWh a year is ElCom's H4 profile. CSCS publish a PUE below 1.2, cooling from Lake Lugano (cscs.ch).